

Assessment 4

This assessment contains two questions that will test your ability to create and package your code in R and Python. Question 1 is on R, while Question 2 is on Python. It should take approximately 8 hours of effort. The available marks are indicated in brackets for each question. Note that this assessment will count **50%** towards the final grade for this module. This assessment will be marked, and feedback will be provided.

Make sure that you carefully read the following Instructions.

Instructions

You are provided with the files `PDS_A4_Q1.Rmd` and `PDS_A4_Q2.ipynb` as well as pdf exports for easier reading. Create files and folders as described in these notebooks. **Marks may be deducted if these layout and format instructions are not followed.** Modify the code files according to the questions contained there, filling in the appropriate cells/code blocks. Please carefully consider the following important points:

- **Do not load any additional libraries in R and Python that are not already explicitly mentioned in the provided notebooks.**
- Answer in each cell/code block only the corresponding subpart (e.g., only answer D in the cell below the heading Part D). The markers will try, where possible, not to penalise answers to parts for errors in previous parts.
- You may use code and variables from previous subparts in your answers of a particular part, e.g., Part D can use code or variables from Part C.
- For Python, provide doc string comments, and for R `roxygen2` style comments (as described in the Blackboard videos of week 9) for every attribute and method you define. You also should add code comments as usual.

Submission

Create a zip file with your college CID containing the following:

- a folder `Q1` with
 - a folder `markovSim` containing R package files (see question 1)
 - a package file `markovSim_0.0.1.tar.gz`
 - the file `PDS_A4_Q1.Rmd` with your answers to question 1D
 - a subfolder `Q1E` containing all the files to answer question 1E
- a folder `Q2` with
 - a folder `metropolisHastingsPackage` containing Python package files (see question 2)
 - a package file `mcmcsampling-0.0.1.tar.gz`
 - the file `PDS_A4_Q2.ipynb` with your answers to question 2D

- a subfolder Q2E containing all the files to answer question 2E

The contents of the zip file can be visualised as follows:

```
CID_PDS_Assessment4.zip
|-- Q1
|   |-- markovSim/
|   |-- markovSim_0.0.1.tar.gz
|   |-- PDS_A4_Q1.Rmd
|   |-- Q1E/
|-- Q2
|   |-- metropolisHastingsPackage/
|   |-- mcmcSampling-0.0.1.tar.gz
|   |-- PDS_A4_Q2.ipynb
|   |-- Q2E/
```

Submit the zip file on Blackboard in the Assessment 4 submission tab on the module page. The deadline is **Monday 15 December 2025 at 09:00am** UK time.

Please note Imperial College's policy on the [late submission of assessments](#).

The assessment must be attempted individually. It must be your own, unaided work. Candidates are prohibited from discussing assessed coursework, and must abide by [Imperial College's rules](#). Enabling other candidates to plagiarise your work constitutes an examination offence. To ensure quality assurance is maintained, departments may choose to invite a random selection of students to an 'authenticity interview'.